SOLUTION ARCHITECTURE & DATA DESIGN

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine

1. Layered Application Design

- Presentation Layer: HTML pages, custom CSS glassmorphism, and Bootstrap 5 responsive elements.
- Controller Layer: Flask endpoints (app.py) that manage forms, invoke models, and log predictions.
- Model Pipeline Layer: Saved pickle (.pkl) files (scalers, encoders, estimators).
- Data Persistence: local CSV history tracker and SQLite database files.

2. Data Schema and Log Structure

User prediction history is written to 'datasets/user_predictions.csv' with the following schema:

Field Name	Data Type	Description
N / P / K	Float	Nutrient levels (kg/ha).
temperature / humidity	Float	Climatic values.
ph / rainfall	Float	Soil pH and rainfall (mm).
crop	String	Recommended crop name.
yield	Float	Expected yield (t/ha).
soil_cluster	Integer	K-Means category ID (0-4).
location / season	String	Region and planting season.

Opticrop - Solution Architecture

Layer 1 - Client UI
HTML5 forms, CSS3 glassmorphism, Bootstrap 5 responsive grid

Layer 2 - Flask Server
/predict POST route: validates inputs -> scales -> infers -> logs to CSV
/dashboard GET route: renders analytics charts

Layer 3 - ML Pipeline
best_model.pkl (RF Classifier)
yield_model.pkl (RF Regressor)
scaler.pkl (StandardScaler)
kmeans_model.pkl(K-Means, k=5)

Layer 4 - Data Store
datasets/user_predictions.csv - append-only prediction log

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